INSECTID

BY MICHELLE GLEESON

Insects are everywhere! They scurry past us in the kitchen, pop up in our gardens, or zoom past us in the park. While we encounter them almost every day, many people don't know a stink bug from a stick insect. This handy identification key will have you classifying insects like an expert in no time!

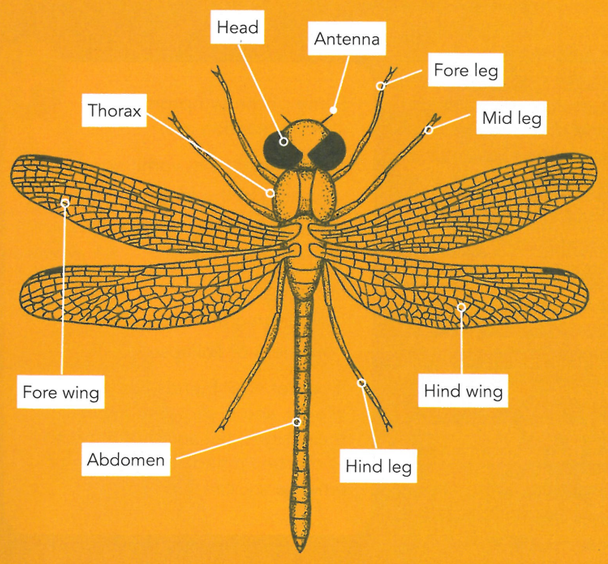
HOW DO WE IDENTIFY INSECTS?

Scientists use tools called identification keys – also known as dichotomous (dye-KOT-oh-moss) keys – to identify things. The key asks you questions about what the insect looks like, discarding possibilities each time until you are left with a single answer. This answer tells you the group, or Order, that the insect belongs to.

HOW TO USE THE IDENTIFICATION KEY

Start at Question 1 and gradually work your way through the identification key. It is a bit like a 'choose your own adventure' book, with the answer for each question leading you to a different part of the key.

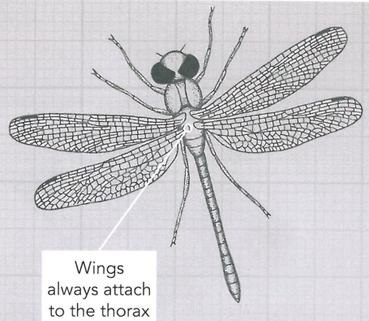
Keep a pen and paper handy and use it to track which path you have taken. That way if you make a mistake, you can simply backtrack, rather than having to start again from the beginning. The diagram below outlines the basic body parts of an insect.



1. Does the insect have wings?

YES → Go To Question 2

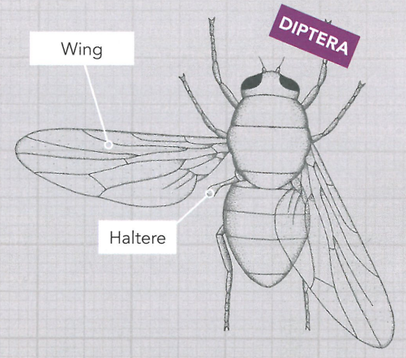
NO → Go To Question 13



2. How many wings does the insect have?

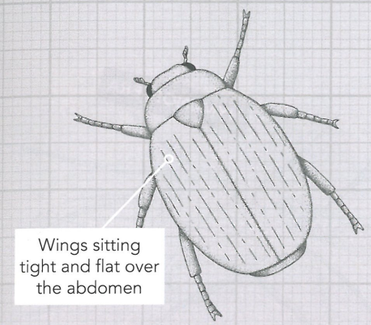
TWO → Answer! Fly, Order Diptera (DIP-ter-ra)

FOUR → Go To Question 3



TIP: HIDDEN WINGS

Some insects keep their delicate hind wings tucked underneath their fore wings, making them difficult to see. If your insect does not look like a fly, then go to Question 3, no matter how many wings you can count.



3. Do the fore wings form a shell-like cover that sits tight and flat over the full length of its abdomen?

YES → Go To Question 4

NO, the wings are either held like a roof over the abdomen; loosely cover the abdomen; sit out away from the body; OR do not reach to the end of the abdomen → Go To Question 5

TIP: SHELL-LIKE WINGS

For insects with fore wings that form shell-like covers, the:

— fore wings are not see-through

— hind wings are usually not visible at all

— wings extend all the way to the end of the abdomen, but not past it.

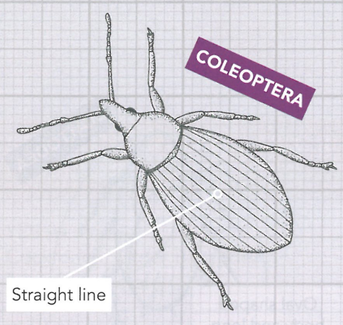
For insects with fore wings that do not form a shell-like cover, you will often see part of the hind wing poking out from under the fore wings.

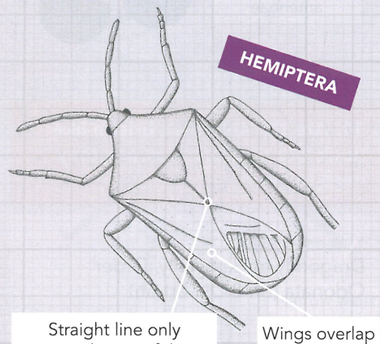


4. How do the wings join?

If the insect's wings join in a straight line that runs down the full length of the abdomen → Answer! Beetle, Order Coleoptera (koll-ee-OP-ter-ra)

If the insect's wings overlap (make a straight line that extends only part of the way down the abdomen), OR if there is no straight line at all → Answer! Stink bug, water bug or plant bug, Order Hemiptera (hem-IP-ter-ra)





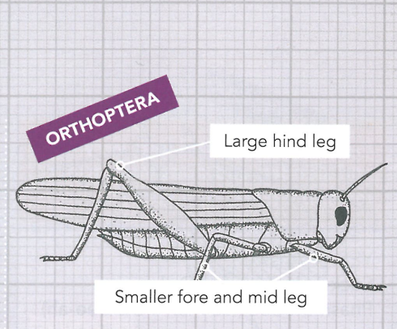
5. Are the wings and body scaly?

YES → Answer! Butterfly or moth, Order Lepidoptera (lep-ee-DOP-ter-ra)

NO → Go To Question 6

TIP: SCALES AREN'T SEE-THROUGH

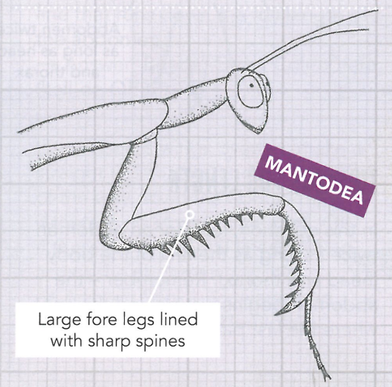
When an insect's wings are covered in scales, you usually can't see through them.



6. Are the hind legs much fatter and look quite different from the other pairs of legs?

YES → Answer! Grasshopper or cricket, Order Orthoptera (or-THOP-ter-ra)

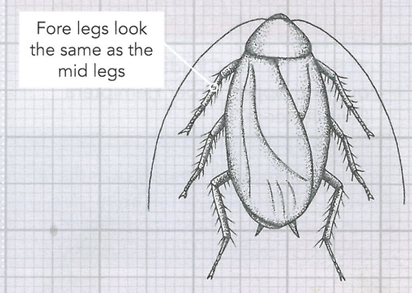
NO, the hind legs look similar to the mid legs → Go To Question 7



7. Are the fore legs large and lined with sharp spines for grabbing prey?

YES → Answer! Praying mantid, Order Mantodea (man-TOE-dee-ah)

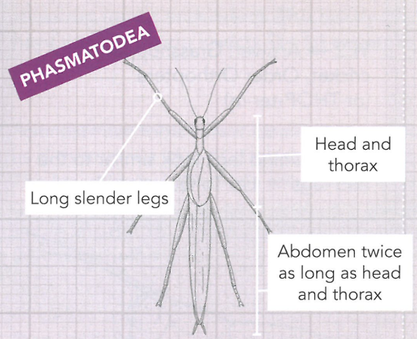
NO, the fore legs look the same as the mid legs → Go To Question 8



8. Are the body and legs very long and slender, and does your insect resemble a leaf or a stick?

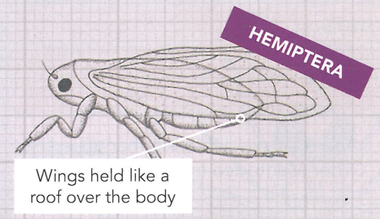
YES → Answer!
Stick or leaf insect, Order Phasmatodea (faz-mah-TOE-dee-ah)

NO → Go To Question 9



TIP: LOOK FOR LENGTH

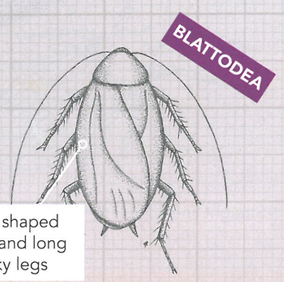
In this question, look for an abdomen twice as long as the rest of the body (head and thorax). Remember – the abdomen begins just behind the spot where the last pair of legs join the body.



9. Are the wings held like a roof over the insect's body?

YES → Answer! Cicada, leafhopper or planthopper, Order Hemiptera (hem-IP-ter-ra)

NO, the wings fold loosely over the body, sit alongside or out away from the body, OR are held up vertically above the body → Go To Question 10



10. Is your insect oval-shaped, with long spiky legs?

YES → Answer!
Cockroach, Order Blattodea (blat-OH-dee-ah)

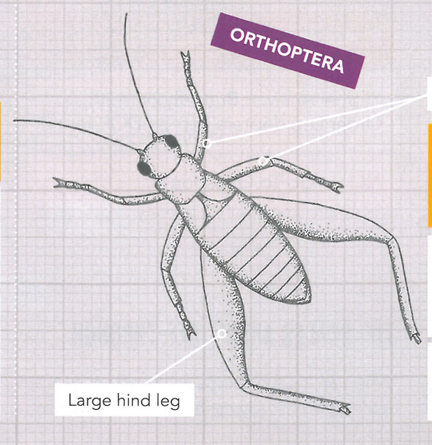
NO → Go To Question 11

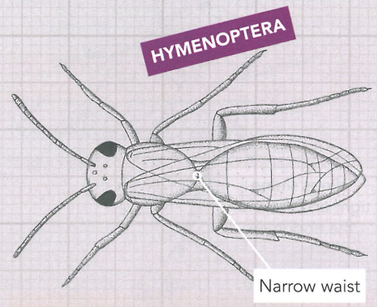


11. Are the antennae bristle-like (look like tiny hairs), or are they hard to see?

YES → Answer!
Dragonfly or damselfly, Order Odonata (oh-DON-art-ah)

NO, the antennae are easy to see → Go To Question 12

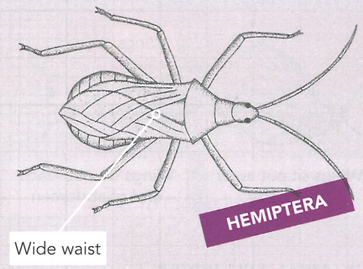




12. Does your insect have a narrow waist?

YES → Answer!
Bee or wasp, Order Hymenoptera (hi-men-OP-ter-ra)

NO → Answer!
Stink bug, water bug or plant bug, Order Hemiptera (hem-IP-ter-ra)



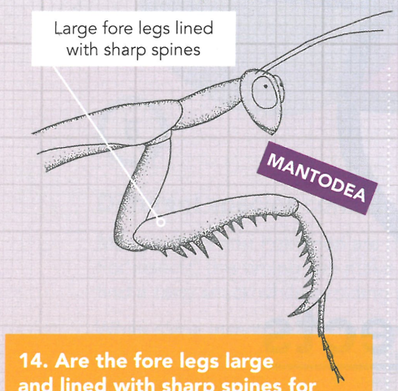
WHAT, A WAIST?

A 'waist' is a very human-sounding characteristic! In an insect, the 'waist' is the join between the thorax and the abdomen.

13. Are the hind legs much fatter and look quite different from the other pairs of legs?

YES → Answer!
Grasshopper or cricket, Order Orthoptera (or-THOP-ter-ra)

NO, the hind legs look similar to the mid legs → Go To Question 14



14. Are the fore legs large and lined with sharp spines for seizing prey?

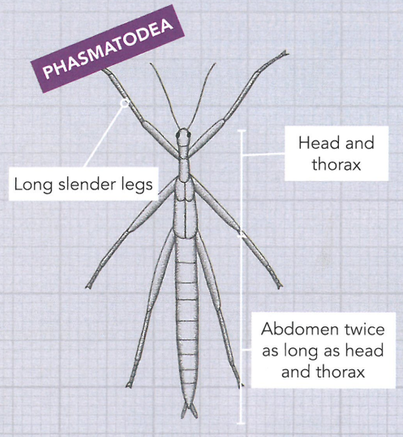
YES → Answer!
Praying mantid, Order Mantodea (man-TOE-dee-ah)

NO, the fore legs look the same as the mid legs → Go To Question 15

15. Are the body and legs very long and slender, and does the insect resemble a leaf or a stick?

YES → Answer!
Stick or leaf insect, Order Phasmatodea (faz-mah-TOE-dee-ah)

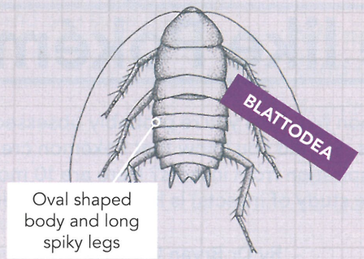
NO → Go To Question 16

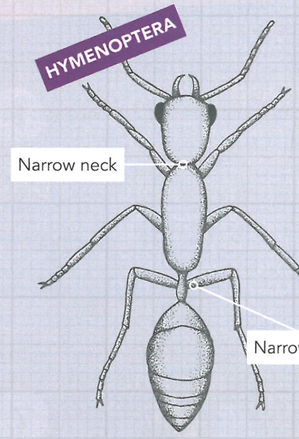


16. Is the insect oval-shaped, with long spiky legs?

YES → Answer!
Cockroach, Order Blattodea (blat-OH-dee-ah)

NO → Go To Question 17

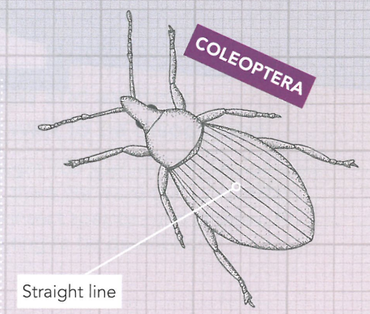




17. Does it have a narrow neck and waist?

YES → Answer!
Ant, Order Hymenoptera (hi-men-OP-ter-ra)

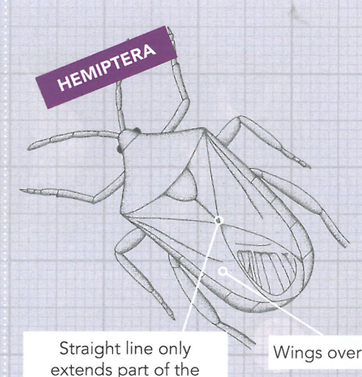
NO → Go To Question 18



18. If you haven't placed the insect into any of the other Orders by now, it will most likely fall into one of the two below. These insects do have wings, but because they are folded tightly over their bodies you may not have noticed them.

If the insect's wings join in a straight line that runs down the full length of the abdomen
→ Answer! Beetle, Order Coleoptera (koll-ee-OP-ter-ra)

If the insect's wings overlap (make a straight line that extends only part of the way down the abdomen), OR if there is no straight line at all, OR if there are horizontal lines running across the abdomen
→ Answer! Stink bug, water bug or plant bug, Order Hemiptera (hem-IP-ter-ra)



This key is adapted from the book *Miniature Lives*, by Michelle Gleeson.

Brought to you by CSIRO Publishing, this book contains simple explanations, amusing stories and quirky facts about insects. It's perfect for those keen on entomology (the study of bugs)! www.publish.csiro.au